

Solution Requirements

Date	15 March 2026
Team ID	-
Project Name	Flood Prediction Using Machine Learning
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure login for Admin and Disaster Management Officers.	High
2	Authorization	Role-based access for Admin and Officers.	High
3	External Interfaces	Integrate weather APIs, rainfall, river level and GIS data.	High
4	Transaction Processing	Collect, preprocess and predict flood risk using ML.	High
5	Reporting	Generate dashboards and PDF/CSV reports.	Medium
6	Business Rules	Classify risk as Low/Moderate/High and send alerts.	High
7	Compliance	Secure storage and privacy compliance.	Medium
8	Other	Real-time alerts and map visualization.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast prediction processing.	<2 seconds
2	Scalability	Handle growing users/data.	10,000+ requests/day
3	Security & Data Privacy	Encryption and secure authentication.	AES-256, HTTPS

4	Reliability & Availability	Continuous service.	99.9% uptime
5	Usability & Accessibility	Responsive easy-to-use interface.	Web Page
6	Maintainability & Portability	Easy updates and deployment.	Windows/Linux compatible